

AUTOMATION PORTFOLIO — TECHNICAL RECORD

SOPHON FINANCE SYSTEMS

AI-Driven Finance & Accounting Automation | github.com/sophonfinance-wq/finance-automation-portfolio

CASH & DEBT RECONCILIATION

A deterministic, materiality-based cash and debt reconciliation engine that ties every account to its bank or lender source by account number and emits an audit-ready, five-section evidence log in Markdown and color-coded Excel.

Report No.	SFS-ENG-002	Package	recon_engine
System	Cash & Debt Reconciliation	Data	Fictional / seeded
Run	python -m recon_engine	Status	Public — portfolio record

1.0 OVERVIEW AND SCENARIO

At month-end close across four fictional entities, one analyst overstates Cedar Ridge Partners LLC's general-ledger term loan (DEBT-2003) by \$12,500 relative to the lender's three-part statement, and a \$1,875.40 deposit clears Maple Fund LP's bank (CASH-1001) before it is booked to the ledger. The engine reconciles all nine seeded accounts by account number, classifies a small \$18.75 interest accrual as immaterial timing noise, and escalates only the two genuinely material breaks as FLAG-001 and FLAG-002 with their likely causes cited, while skipping a dormant zero-activity escrow with a documented note. Separately, the assurance loop catches a drifted package in which an active account was mis-marked dormant and silently dropped from scope, restoring it to the source of record before the package ships.

2.0 WHAT THE SYSTEM AUTOMATES

- Reconciliation of multiple trial-balance accounts (four cash, four debt across four fictional entities) on a repeatable monthly cadence
- Cash tie-out of GL cash balance against the bank statement ending balance, matched by account number rather than row position
- Debt tie-out of GL loan balance against a lender three-part total (principal + current interest/reserve + late paydown)
- Signed variance computation, materiality classification (clean / timing / flag), and sequential FLAG-### numbering
- Skipping of dormant zero-activity accounts with a documented policy note instead of chasing zeros
- Root-cause annotation of each flagged line, citing the deposit-in-transit or keying-error explanation
- Generation of a five-section Markdown evidence log plus a color-coded five-sheet Excel workbook
- Drift remediation via the assurance loop, which restores a mutated package to the seeded source of record one account at a time

3.0 OPERATING PROCEDURE

- 1 Generate the seeded synthetic dataset (GL cash and debt, bank statements, lender statements for four entities) and inject five known discrepancies deterministically

- 2 Index bank and lender source statements by account number
- 3 Iterate GL records, skipping dormant or zero-activity rows with a documented note
- 4 Reconcile cash (GL vs bank ending) and debt (GL vs lender three-part total) for each in-scope account
- 5 Compute the signed GL-minus-source variance rounded to cents and classify it clean / timing / flag against the materiality threshold
- 6 Assign sequential FLAG-### identifiers to escalated lines in a stable cash-then-debt order
- 7 Render the five-section evidence log to Markdown and to a color-coded five-sheet .xlsx workbook via openpyxl
- 8 Print a console summary with counts and flagged lines; return exit code 0 (flags are an expected outcome, not a failure)
- 9 Optionally run the assurance loop: observe, detect deviations from the pristine baseline, resync the lowest-numbered drifted account, re-verify, and apply a materiality gate (PASS / FLAG / FAIL)

4.0 CONTROLS AND SAFEGUARDS

- Deterministic seeded generation (default seed 20260331) yields byte-identical, fully reproducible runs for the same inputs
- Values targeted by account number, never by row index, which defeats off-by-one row drift; the assurance loop additionally records each overwritten field as a booked correction
- Materiality threshold (default \$50.00) separates immaterial timing noise, which is commented, from material breaks, which become numbered flags escalated for review
- Independent re-derivation in the assurance loop compares every account against the reconciliation of the pristine seeded source rather than assuming an all-clean target, so genuine reconciling items survive and are not zeroed out
- Separation of duties between source of record and remediation: the seeded generator is the system of record and every loop correction is a logged restoration, never an invented number
- Standardized evidence trail in which each value carries source, variance, and status, and close-out separates timing items from structural items
- Materiality-gated verdict mapped to process exit codes (FAIL returns 1, PASS and FLAG return 0) so a non-converged package can block downstream steps

5.0 VERIFICATION AND TESTING

Correctness is proven by a suite of 1,011 automated pytest tests that all pass, covering variance arithmetic, classification boundaries around the materiality threshold, the three-part debt formula, each injected discrepancy, Markdown and Excel rendering, the CLI, and the assurance loop. The deterministic seed makes every run reproducible, and the assurance loop proves convergence by re-deriving each account against the pristine baseline. Running the engine and the loop demo reproduced the documented output exactly, including the two flags (FLAG-001 CASH-1001 at -1,875.40 and FLAG-002 DEBT-2003 at 12,500.00) and the loop's three-turn convergence with a FLAG verdict.

6.0 KEY FACTS

Attribute	Value
Automated tests passing	1,011

Accounts in scope	9 (4 cash, 4 debt, 1 dormant skipped)
Default materiality threshold	\$50.00
Evidence-log sections	5 (Summary, Cash, Debt, Flagged, Notes)
Seeded discrepancies injected	5 (deposit-in-transit, timing, keying error, late-paydown, dormant)
Runtime dependency	openpyxl only (standard library otherwise)

This record documents a self-contained system in the Sophon Finance Systems automation portfolio. All data is fictional and seeded; no employer or client workpaper, entity, methodology, path, or figure is reproduced. Prepared 14 July 2026.